

STELLWAGEN BUOY 44029, MA, USA

WMO: 997140

Lat: 42.523N

Lon: 70.566W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (NAE)

Period: 06-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.5	16.7	N/A	N/A	N/A	N/A	N/A	N/A	38.3	25.8	36.0	24.4	22.7	310	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	74.3	N/A	72.8	N/A	71.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.5	29.6	25.0	DB	7.1	77.2	2.6	2.7	5.2	79.2	3.7	80.8	2.2	82.3	0.4	84.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	31.6	31.6	35.8	43.9	52.3	60.9	67.5	67.5	63.3	54.7	46.0	39.0
	DBStd	13.96	8.57	6.96	5.97	4.29	4.36	4.48	3.13	2.69	3.95	5.06	6.15	6.94
	HDD50	2246	571	516	440	189	23	1	0	0	0	19	145	342
	HDD65	5833	1036	936	905	634	394	135	11	9	78	318	571	806
	CDD50	2103	0	0	1	5	95	328	542	542	399	166	24	1
	CDD65	214	0	0	0	0	1	12	88	86	27	0	0	0
	CDH74	62	0	0	0	0	0	4	44	9	5	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	13.2	17.6	16.9	15.0	12.6	10.9	10.2	8.9	9.3	11.1	14.1	15.3	16.5
Precipitation	PrecAvg	47.6	3.9	3.5	4.4	4.1	3.6	3.9	3.3	3.5	4.0	4.5	4.0	4.5
	PrecMax	65.7	6.5	8.5	16.6	10.1	16.8	11.3	8.5	8.0	8.8	13.3	8.0	7.3
	PrecMin	33.9	1.1	1.6	0.7	0.4	0.6	0.5	1.2	1.0	1.6	0.2	1.1	1.5
	PrecStd	7.6	1.2	1.5	3.2	2.2	3.0	2.5	1.9	1.9	2.0	3.2	1.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.9	46.8	53.0	57.5	66.9	73.5	77.7	75.3	74.4	66.7	59.8	54.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	47.6	44.4	47.4	53.5	62.4	70.0	75.1	73.4	72.1	64.2	56.5	51.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	45.6	42.3	44.7	51.1	60.3	68.0	73.3	72.3	70.4	62.4	55.0	50.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	43.4	40.9	42.6	49.4	58.4	66.3	71.8	71.3	68.6	61.0	53.7	48.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	9.6	9.4	7.7	5.9	5.9	5.5	5.7	5.3	5.1	6.1	6.6	8.5
		MCDBR	9.9	9.1	8.7	8.3	8.8	7.3	7.3	6.2	6.1	6.0	6.6	9.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.297	0.304	0.324	0.367	0.403	0.452	0.470	0.434	0.378	0.337	0.313	0.298	
	taud	2.530	2.521	2.512	2.418	2.361	2.257	2.211	2.303	2.442	2.550	2.577	2.577	
	Ebn at Noon	271	288	293	286	277	263	256	262	271	271	263	262	
	Edh at Noon	22	26	30	35	38	43	44	39	32	25	21	20	
All-Sky Solar Radiation	RadAvg	534	811	1137	1474	1659	1786	1885	1695	1352	900	596	442	
	RadStd	36	71	78	159	178	161	119	64	86	74	46	51	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (68 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+76	(47)

Nomenclature: See separate page